INCH-POUND
J-W-1177/2B
June 10, 1988
SUPERSEDING
J-W-1177/2A
September 27, 1976

FEDERAL SPECIFICATION SHEET

# WIRE, MAGNET, ELECTRICAL, CLASS 105, TYPE SU, SOLDERABLE POLYURETHANE COATED, ROUND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of J-W-1177.

Classification: Class 105; type SU (single), type SU2 (heavy); round.
Insulating materials: The film shall be based on a solderable polyurethane resin.
NEMA/ANSI equivalent: All test requirements except thermal endurance are equivalent to MW-2 of NEMA MW 1000.
General requirements: See J-W-1177 for general requirements, quality assurance provisions, and packaging.

Requirements:

| Characteristics | Test procedure, see J-W-1177 | Wire sizes, AWG | Requirements |
|---|---|---|---|
| Dimensions | 4.7.1.2 | 25-56 | See tables I and II. |
| Adherence and flexibility | 4.7.2.1 | 25-56 | No cracks visible in the film coating. |
| Elongation | 4.7.5 | 25-50 | Not less than the value in table III. |
| Heat shock | 4.7.4 | 25-44 | No cracks visible in the coating after conditioning as shown in table IV. |
| Scrape resistance | 4.7.6 | 25-30 | Lowest grams-to-fail load for any of the three tests and the average of the three tests shall be not less than the values in table V. |
| Springback | 4.7.7 | 25-30 | Not greater than the value in table VI. |
| Dielectric strength | 4.7.9 | 25-44 | Not less than the value in table VII. |

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Requirements: (Continued)

| Characteristics | Test procedure, see J-W-1177 | Wire sizes, AWG | Requirements |
|---|---|---|---|
| Continuity | 4.7.10 | 31-56 | The number of discontinuities shall be not greater than the number listed in table VIII. |
| | 4.7.11 | 25-30 | |
| Thermoplastic flow | 4.7.8 | 36 | Median not less than 170°C with heavy film coated wire. |
| Solubility | 4.7.12 | 36 | Heavy film coated wire shall not soften sufficiently to expose bare conductor when immersed in xylene. |
| Dielectric strength at temperature | 4.7.14 | 36 | Heavy film coated wire shall average not less than 1900 volts. |
| Thermal endurance | 4.7.15.1 | 18 | 105°C minimum with heavy film coated wire. |
| | 4.7.15.2 | 25-44 | 1000 volts/mil minimum after 168 hours at 180°C. |
| | 4.7.15.3 | 25-44 | 150°C minimum as shown in table IV. |
| Solderability | 4.7.17 | 25-46 | Covered with continuous film of solder and not readily separable after soldering as shown in table IX. |

TABLE I. Dimensions, sizes 25 to 44 AWG.

| AWG size | Bare wire diameter, inch | | | Type SU, single | | Type SU2, heavy | |
|---|---|---|---|---|---|---|---|
| | Minimum | Nominal | Maximum | Minimum increase in diameter, inch | Maximum overall diameter, inch | Minimum increase in diameter, inch | Maximum overall diameter, inch |
| 1/25 | 0.0177 | 0.0179 | 0.0180 | 0.0009 | 0.0194 | 0.0018 | 0.0203 |
| 1/26 | .0157 | .0159 | .0160 | .0009 | .0173 | .0017 | .0182 |
| 27 | .0141 | .0142 | .0143 | .0008 | .0156 | .0016 | .0164 |
| 28 | .0125 | .0126 | .0127 | .0008 | .0140 | .0016 | .0147 |
| 29 | .0112 | .0113 | .0114 | .0007 | .0126 | .0015 | .0133 |
| 30 | .0099 | .0100 | .0101 | .0007 | .0112 | .0014 | .0119 |
| 31 | .0088 | .0089 | .0090 | .0006 | .0100 | .0013 | .0108 |
| 32 | .0079 | .0080 | .0081 | .0006 | .0091 | .0012 | .0098 |
| 33 | .0070 | .0071 | .0072 | .0005 | .0081 | .0011 | .0088 |
| 34 | .0062 | .0063 | .0064 | .0005 | .0072 | .0010 | .0078 |
| 35 | .0055 | .0056 | .0057 | .0004 | .0064 | .0009 | .0070 |
| 36 | .0049 | .0050 | .0051 | .0004 | .0058 | .0008 | .0063 |
| 37 | .0044 | .0045 | .0046 | .0003 | .0052 | .0008 | .0057 |

See footnote at end of table.



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TABLE I. Dimensions, sizes 25 to 44 AWG. - Continued

| AWG size | Bare wire diameter, inch | | | Type SU, single | | Type SU2, heavy | |
|---|---|---|---|---|---|---|---|
| | Minimum | Nominal | Maximum | Minimum increase in diameter, inch | Maximum overall diameter, inch | Minimum increase in diameter, inch | Maximum overall diameter, inch |
| 38 | 0.0039 | 0.0040 | 0.0041 | 0.0003 | 0.0047 | 0.0007 | 0.0051 |
| 39 | .0034 | .0035 | .0036 | .0002 | .0041 | .0006 | .0045 |
| 40 | .0030 | .0031 | .0032 | .0002 | .0037 | .0006 | .0040 |
| 41 | .0027 | .0028 | .0029 | .0002 | .0033 | .0005 | .0036 |
| 42 | .0024 | .0025 | .0026 | .0002 | .0030 | .0004 | .0032 |
| 43 | .0021 | .0022 | .0023 | .0002 | .0026 | .0004 | .0029 |
| 44 | .0019 | .0020 | .0021 | .0001 | .0024 | .0004 | .0027 |

1/ These bare wire diameters may be exceeded, provided:

(a) The maximum diameters specified by QQ-W-343 are not exceeded,
(b) The minimum increases in diameter shown in table I are maintained, and
(c) The maximum overall diameters shown in table I are not exceeded.

TABLE II. Characteristics of wire sizes 45 to 56 AWG.

| AWG size | Theoretical 1/ nominal bare wire diameter, inch | Conductor resistance at 20°C, ohms per foot | | | Type SU, single | | Type SU2, heavy | |
|---|---|---|---|---|---|---|---|---|
| | | Minimum | Nominal | Maximum | Minimum increase in diameter, inch | Maximum overall diameter, inch | Minimum increase in diameter, inch | Maximum overall diameter, inch |
| 45 | 0.00176 | 3.080 | 3.348 | 3.616 | 0.00010 | 0.00205 | 0.00030 | 0.00230 |
| 46 | .00157 | 3.870 | 4.207 | 4.544 | .00010 | .00185 | .00030 | .00210 |
| 47 | .00140 | 4.868 | 5.291 | 5.714 | .00010 | .00170 | .00030 | .00190 |
| 48 | .00124 | 6.205 | 6.745 | 7.285 | .00010 | .00150 | .00020 | .00170 |
| 49 | .00111 | 7.744 | 8.417 | 9.090 | .00010 | .00130 | .00020 | .00150 |
| 50 | .00099 | 9.734 | 10.58 | 11.43 | .00010 | .00120 | .00020 | .00140 |
| 51 | .00088 | 12.32 | 13.39 | 14.46 | .00010 | .00110 | ------- | ------- |
| 52 | .00078 | 15.69 | 17.05 | 18.41 | .00010 | .00100 | ------- | ------- |
| 53 | .00070 | 19.48 | 21.17 | 22.86 | .00005 | .00085 | ------- | ------- |
| 54 | .00062 | 24.82 | 26.98 | 29.14 | .00005 | .00075 | ------- | ------- |
| 55 | .00055 | 31.54 | 34.28 | 37.02 | .00005 | .00070 | ------- | ------- |
| 56 | .00049 | 39.73 | 43.19 | 46.65 | .00005 | .00065 | ------- | ------- |

1/ Theoretical nominal bare wire diameters are in accordance with NBS Handbook 100. Conductor diameter tolerances are shown as resistance values and shall be determined by measuring the resistance of the wire in accordance with ASTM B 193, where applicable. A specimen at least 5 feet long shall be used.



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TABLE III. Elongation of finished wire.

| AWG size | Minimum elongation, percent | AWG size | Minimum elongation, percent |
|---|---|---|---|
| 25 | 28 | 38 | 19 |
| 26 | 27 | 39 | 18 |
| 27 | 27 | 40 | 17 |
| 28 | 26 | 41 | 17 |
| 29 | 26 | 42 | 16 |
| 30 | 25 | 43 | 15 |
| 31 | 24 | 44 | 14 |
| 32 | 24 | 45 | 11 |
| 33 | 23 | 46 | 10 |
| 34 | 22 | 47 | 8 |
| 35 | 21 | 48 | 7 |
| 36 | 20 | 49 | 6 |
| 37 | 20 | 50 | 5 |

TABLE IV. Heat shock.

| AWG size | Minimum elongation, percent | Mandrel diameter | Minimum temperature, °C |
|---|---|---|---|
| 25-30 | 20 | 6X | 150 |
| 31-44 | 1/ 20 | 6X | 150 |

1/ Or to the breaking point, whichever is less.

TABLE V. Scrape resistance.

| AWG size | Type SU, single | | Type SU2, heavy | |
|---|---|---|---|---|
| | Average grams-to-fail | Minimum grams-to-fail | Average grams-to-fail | Minimum grams-to-fail |
| 25 | 350 | 300 | 720 | 615 |
| 26 | 330 | 280 | 675 | 575 |
| 27 | 310 | 265 | 635 | 540 |
| 28 | 290 | 250 | 595 | 510 |
| 29 | 270 | 230 | 560 | 480 |
| 30 | 255 | 220 | 525 | 450 |



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TABLE VI. Springback of finished wire.

| AWG size | Maximum springback, degrees |
|---|---|
| 25 | 72 |
| 26 | 76 |
| 27 | 50 |
| 28 | 55 |
| 29 | 61 |
| 30 | 66 |

TABLE VII. Minimum breakdown voltages.

| AWG size | Volts | | AWG size | Volts | |
|---|---|---|---|---|---|
| | Type SU | Type SU2 | | Type SU | Type SU2 |
| 25 | 2625 | 4725 | 35 | 1325 | 2750 |
| 26 | 2550 | 4600 | 36 | 1200 | 2525 |
| 27 | 2500 | 4500 | 37 | 1075 | 2325 |
| 28 | 2425 | 4375 | 38 | 950 | 2150 |
| 29 | 2375 | 4250 | 39 | 850 | 1975 |
| 30 | 2300 | 4150 | 40 | 775 | 1800 |
| 31 | 2075 | 3825 | 41 | 700 | 1675 |
| 32 | 1850 | 3525 | 42 | 625 | 1525 |
| 33 | 1675 | 3250 | 43 | 550 | 1400 |
| 34 | 1500 | 2975 | 44 | 500 | 1300 |

TABLE VIII. Continuity.

| AWG size | Maximum number of discontinuities | |
|---|---|---|
| | Type SU | Type SU2 |
| 25-30 | 25 | 7 |
| 31-46 | 25 | 5 |
| 47-50 | 25 | 10 |
| 51-56 | 25 | — |

TABLE IX. Solderability.

| AWG size | Maximum immersion time, seconds | Temperature of solder, °C |
|---|---|---|
| 25-29 | 6 | 360 |
| 30-36 | 5 | 360 |
| 37-46 | 4 | 360 |



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Part number: Magnet wire covered by this specification shall be defined by the following part numbering system.
Example: M1177/2-02C029

| M1177/2- | 02 | C | 029 |
|---|---|---|---|
| Federal specification identifier | Two digit type code | Single letter conductor code | Three character size code |

The following codes shall apply:

| Type | Type code | Conductor | Conductor code |
|---|---|---|---|
| SU | 01 | Copper | C |
| SU2 | 02 | Aluminum | A |
| | | Nickel-coated copper | N |
| | | Silver-coated copper | S |

The size code shall be the bare wire dimension. AWG wire size shall be used.

Intended use: Type SU magnet wire is intended for use in 105°C applications similar to those for which type I is used and where a solerable wire is desired.

Revision letters are not used to denote changes due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AR, ER, MI
DLA - IS

User activities:
Army - ME
Navy - AS, CG, MC, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PBO, PCD
INTERIOR - BLM
HHS - FDA
DCGOVT - DCG
NASA - JFK
COMMERCE - NBS
TRANSPORTATION - APM, FAA

Preparing activity:
Navy - SH
(Project 6145-1111-01)



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